

MATLAB: Session 5

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23 October 2025

Exercises

Part A – Linear Applications and Change of Basis

In the following, $\mathcal{E}_n = \{e_1, \dots, e_n\}$ denotes the canonical basis for $\mathbb{R}^n$

A.1 (*From last time*) Consider the endomorphism of $\mathbb{R}^3$, $s_m : \mathbb{R}^3 \rightarrow \mathbb{R}^3$, described as

$$\begin{cases} s_m(e_1) = me_1 + e_2 \\ s_m(e_2) = 4e_1 + me_2 \\ s_m(e_3) = -e_1 - e_2 + 2e_3 \end{cases},$$

where m is a real parameter.

A.1.1 Assign the matrix $M_{\mathcal{E}_3, \mathcal{E}_3}(s_m)$ to the variable `E_m`.

A.1.2 Compute the dimension of $\ker s_m$ for each m , and assign it to the variables `dimkers_m`.

A.1.3 Compute the dimension of $\text{Im} s_m$ and assign it to the variables `dimims_m`.

A.1.4 Select an m for which s_m is not surjective. Substitute this m to `E_m`, defining a new variable `E`.

A.1.5 For such an m , compute a basis for $\ker s_m$ and a basis for $\text{Im} s_m$ and assign them to the variables `basiskers` and `basisims`, respectively.

A.2 Let f be the endomorphism of $\mathbb{R}^3$, represent by the matrix

$$A = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 3 \end{bmatrix},$$

on the canonical basis $\mathcal{E}_3$.

A.2.1 Assign the matrix A to the variable **A**.

A.2.2 Let $\mathcal{B} = \{(1, 1, 0), (-1, 0, 1), (1, 1, 1)\}$. Compute the transition matrix from $\mathcal{B}$ to $\mathcal{E}_3$. Assign it to the variable **I_BE3**.

A.2.3 Compute the transition matrix from $\mathcal{E}_3$ to $\mathcal{B}$. Assign it to the variable **I_E3B**.

A.2.4 Compute $M_{\mathcal{B},\mathcal{B}}(f)$. Assign it to the variable **Mf_B**.

A.3 Let $g : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ be the linear map defined as

$$g(x, y) = (x, x, y).$$

A.3.1 Assign the matrix $M_{\mathcal{E}_2, \mathcal{E}_3}(g)$ to the variable **B**.

A.3.2 Let $\mathcal{C} = \{(1, 1, 0), (0, 1, 1), (0, 0, 1)\}$. Compute the transition matrix from $\mathcal{C}$ to $\mathcal{E}_3$. Assign it to the variable **I_CE3**.

A.3.3 Compute the transition matrix from $\mathcal{E}_3$ to $\mathcal{C}$. Assign it to the variable **I_E3C**.

A.3.4 Compute $M_{\mathcal{E}_2, \mathcal{C}}(g)$. Assign it to the variable **Mg_E2C**.

A.4 Let $h : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be the endomorphism represented, with respect to the canonical basis $\mathcal{E}_3$, by the following matrix

$$C = \begin{bmatrix} -1 & 1 & 2 \\ 3 & 3 & 4 \\ 2 & 1 & 1 \end{bmatrix}.$$

A.4.1 Assign the matrix C to the variable **C**.

A.4.2 Compute $M_{\mathcal{D}, \mathcal{D}}(h)$, where $\mathcal{D} = \{(1, 1, -1), (1, 0, 1), (1, 1, 0)\}$. Assign it to the variable **Mh_D**.

A.4.3 Compute the dimension of $\ker h$ and assign it to the variable **dimkerh**.

A.4.4 Compute the dimension of $\text{Im} h$ and assign it to the variable **dimimh**.

A.4.5 Compute a basis for $\ker h$ and assign it to the variable **basiskerh**.

A.4.6 Compute a basis for $\text{Im} h$ and assign it to the variable **basisimh**.

A.4.7 Is h injective? Is h surjective?

Part B – Determinant

B.1 Consider the matrix

$$M = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 3 \\ 2 & 1 & 0 \end{bmatrix}.$$

Assign it to the variable `M`.

B.2 Compute, using Laplace's rule, the determinant of M . Assign it to the variable `detM` and check your result using the MATLAB function `det()`.

B.3 Define the matrix N in the following way: take M , add to its second row two times the first row, then subtract from the third row the sum of the first two rows, finally divide the first row by 17. Assign it to the variable `N`.

B.4 Compute $\det(N)$. Assign it to `detN` and compare it with `detM`. What do you notice?

B.5 Define the matrix L following the steps in **B.3**, but working with columns instead of rows. Assign it to the variable `L`.

B.6 Compute $\det(L)$. Assign it to `detL` and compare it with `detM`. What do you notice?

B.7 Consider the matrix P , obtained from M by swapping the first two rows. Assign it to the variable `P`.

B.8 Compute $\det(P)$. Assign it to `detP` and compare it with `detM`. What do you notice?